

COVID19 on Welfare: a microsimulation approach
(preliminary draft)

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1. About household welfare, COVID19 can generate a set of multiple shocks, as the reduction in labor supply, the increase in prices, and the decrease in remittances. Also, the starting time of these shocks, their durations, and their extents can vary, and then affect households differently.
2. Given the complexity of the nature of COVID19 shocks, the suggested microsimulation tries to synthesize the overall impact and to quantify the effect attributable to the channels listed above.
3. Formally, we assume that the impact of the COVID19 shocks on welfare is approximated by aggregating the effect of the main impact channels. If we denote the welfare of household h at time t_0 by w_{h,t_0} , the change in welfare due to the COVID19 shocks is:

$$\Delta w_{h,t_0,1} = w_{h,t_0} \sum_{j=1}^J (\Delta F_{j,t_0,1})$$

where $\Delta w_{h,t_0,1}$ denotes the change in the welfare of household h between period t_0 (pre-COVID10) and t_1 (after COVID19). $\Delta F_{j,t_0,1}$ is the relative (%) change in factor j between t_0 and t_1 . The two usual proxies of household welfare are per capita expenditures and per capita income, depending on the country. The factors that are expected to affect welfare are remittances, labor incomes (distinguished by economic sector/by source/etc.), and public transfers, in addition to consumer prices (where relevant in such a short time as the one we study).

4. *Period of analysis and period of interest:* since the intensity of changes in the different factors can vary over time, it may be helpful to assess the impact in the shorter term, in the longer term, or even periodically. Based on this, we need to determine the time unit of welfare (e.g., monthly revenue), as well as the time length of interest (e.g., the impact between March and September).
5. Remember that (monetary) welfare depends on two main factors, which are consumer prices and the disposable (nominal) income.
6. Consumer prices are tightly related to producer prices. In addition, consumer prices are related to the fluidity in the distribution and the commercialization of goods. For producers, a rise in the cost of inputs increases their prices. For the distribution and commercialization chains, any turbulence that causes a partial stock shortage reduces the supply and increases prices. Note that some prices may even decrease with the COVID19, like those of the energetic products and, potentially, durable goods.

7. *Price variation and welfare*: To estimate the welfare impact of the (consumer) price variation, we can use the first-order approximation of Taylor, namely:

$$C_{good_prices,h} = \Delta w_h = -e_{good,h} \Delta p_{food}$$

Where $e_{good,h}$ represents the expenditure of household h on the good of interest.

In the presence of multiple goods, one can use a price index deflator, as the Laspeyres index.

8. Household disposable income depends on a set of factors. For instance, assume that the household disposable income includes:
- i. earned net income by each active household member, by economic sector;
 - ii. household internal and external remittances;
 - iii. public transfers (including those existing before COVID19 and those introduced after COVID19);
 - iv. Etc.

To assess the welfare impact, we need:

- i. An estimate of the earned income in each sector and for each active household member. Usual household surveys contain information on whether the household member is (labor) active and in which economic sector he/she works. Some HH surveys contain information on earned income. Otherwise, we will suggest some models to estimate the potential revenues starting from the household expenditures. Note that, in some cases, we can find a large number of non-reported incomes in the survey. For this purpose, it is suggested to treat the missing values with one of the popular methods.

Approaches to replace missing values:

- A- Prediction from a simple linear model (OLS). Disadvantages and Advantages:
 - a. Simple model;
 - b. In general, generates a lot of negative predicted values;
 - c. Reduces significantly the variance of income, since the residuals are not considered.
- B- Prediction from a Tobit II (0) model. Disadvantages and Advantages:
 - a. Does not generate negative predictors;
 - b. Reduces significantly the variance of income, since the residuals are not considered.
- C- Hotdeck model and breakdown by percentiles.
 - a. Simple to do;
 - b. Does not generate negative predictors;
 - c. Preserves income variance.

- ii. An estimate of the change in revenues for each sector (also distinguished by formal and informal). This information can be estimated based on secondary data.

9. *Labor income variation*: the assumption is that the labor/salary variation affects only households with active members and non-zero labor income. If we denote the average household labor income in households h operating in sector s at time t as $r_{h,s,t}$, and the number of active members working in sector s as $n_{h,s,t}$, then the total household labor income is:

$$R_{h,t} = \sum_s n_{h,s,t} r_{h,s,t}$$

And its change with respect to time t_0 is:

$$\Delta R_{h,t_0,1} = \sum_s n_{h,s,t} \Delta r_{h,s,t_0,1}$$

Note that $n_{h,s,t}$ can change over time as well if, for example, workers move from one economic sector to another, or become unemployed, between the two periods. To estimate such changes, we calibrate the total number of workers in sector s at time t according to the official statistics on employment (or estimates) and based on individual probabilities of working in sector s (“job queuing” approach). In other words, if total workers in sector s decreased by 10% due to COVID19, then the 10% of workers in sector s with the lowest probability of being in that sector are moved to the unemployed status.